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НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ  
ТОМСКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ (НИ ТГУ)  
Институт прикладной математики и компьютерных наук

### ОТЧЁТ ПО ЛАБОРАТОРНОЙ РАБОТЕ №3

по дисциплине «Основы построения защищённых компьютерных сетей»

«Идентификация служб и приложений»

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1. С помощью сканера Nessus проведем сканирование адреса 172.16.8.11 портов 1-200

← Back to My Scans

Scan Summary Hosts 1 Vulnerabilities 24 Remediations 1 Notes 2 History 1

Filter Search Vulnerabilities 24 Vulnerabilities

| <input type="checkbox"/> | Sev   | CVSS  | VPR | NaFamily            | Count |   |   |
|--------------------------|-------|-------|-----|---------------------|-------|---|---|
| <input type="checkbox"/> | INFO  | ...   | ... | 2 Port scanners     | 6     | ⊙ | / |
| <input type="checkbox"/> | MIXED | ...   | ... | 3 DNS               | 4     | ⊙ | / |
| <input type="checkbox"/> | MIXED | ...   | ... | 4 Web Servers       | 4     | ⊙ | / |
| <input type="checkbox"/> | MIXED | ...   | ... | 3 Misc.             | 3     | ⊙ | / |
| <input type="checkbox"/> | INFO  |       |     | SeService detection | 3     | ⊙ | / |
| <input type="checkbox"/> | INFO  | ...   | ... | 2 DNS               | 2     | ⊙ | / |
| <input type="checkbox"/> | INFO  | ...   | ... | 2 Misc.             | 2     | ⊙ | / |
| <input type="checkbox"/> | INFO  | ...   | ... | 2 Service detection | 2     | ⊙ | / |
| <input type="checkbox"/> | LOW   | 2.1 * | 2.2 | ICGeneral           | 1     | ⊙ | / |

Scan De  
Policy:  
Status:  
Severity  
Scanner:  
Start:  
End:  
Elapsed:

Vulnera

|                          |      |                     |   |   |   |
|--------------------------|------|---------------------|---|---|---|
| <input type="checkbox"/> | INFO | ApWeb Servers       | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | CoGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | DeGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | EtGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | FTService detection | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | NcSettings          | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | OSGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | OSGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | OSSettings          | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | PaGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | SSGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | TaSettings          | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | TCGeneral           | 1 | ⊙ | / |
| <input type="checkbox"/> | INFO | TrGeneral           | 1 | ⊙ | / |

2. Проанализируем отчёт и определим, какие службы были найдены и идентифицированы

| Port 21/tcp was found to be open                |             |
|-------------------------------------------------|-------------|
| To see debug logs, please visit individual host |             |
| Port                                            | Hosts       |
| 21 / tcp / ftp                                  | 172.16.8.11 |

| Port 22/tcp was found to be open                |             |
|-------------------------------------------------|-------------|
| To see debug logs, please visit individual host |             |
| Port                                            | Hosts       |
| 22 / tcp / ssh                                  | 172.16.8.11 |

| Port 25/tcp was found to be open                |             |
|-------------------------------------------------|-------------|
| To see debug logs, please visit individual host |             |
| Port                                            | Hosts       |
| 25 / tcp                                        | 172.16.8.11 |

| Port 53/tcp was found to be open                |             |
|-------------------------------------------------|-------------|
| To see debug logs, please visit individual host |             |
| Port                                            | Hosts       |
| 53 / tcp / dns                                  | 172.16.8.11 |

| Port 80/tcp was found to be open                |             |
|-------------------------------------------------|-------------|
| To see debug logs, please visit individual host |             |
| Port                                            | Hosts       |
| 80 / tcp / www                                  | 172.16.8.11 |

Найденные службы:

21/tcp - FTP

22/tcp - SSH

25/tcp - найден, но идентифицирован

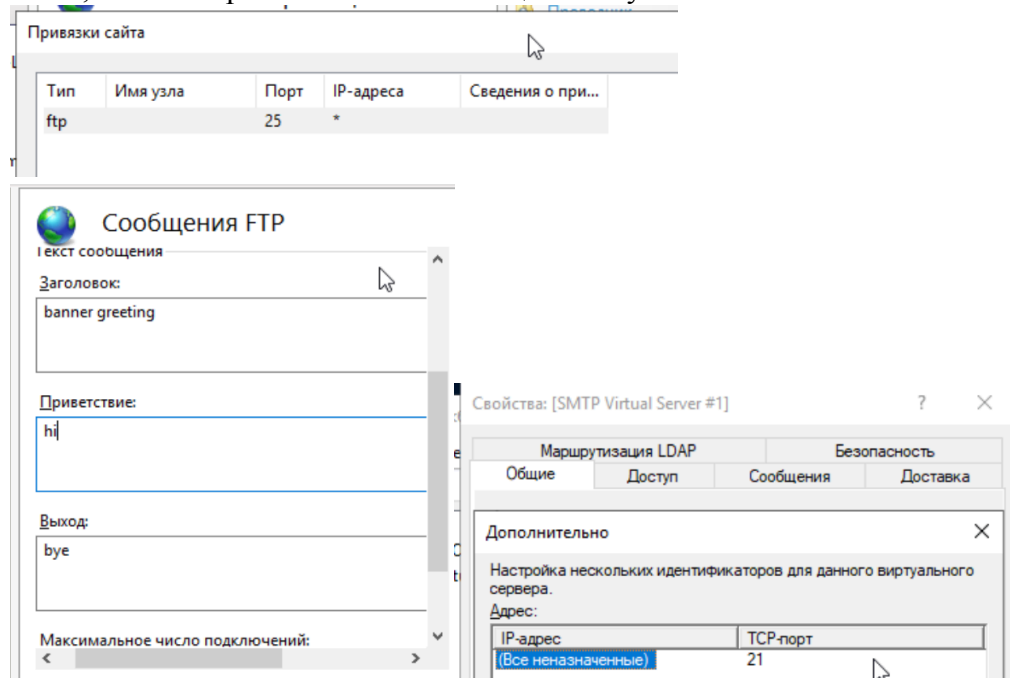
53/tcp - DNS

80/tcp - WWW

| Output                                                                                                                                                      |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| The UDP port scan could not complete: The remote host has remained silent for too long<br>This might be due to a firewall filtering UDP and/or ICMP packets |             |
| To see debug logs, please visit individual host                                                                                                             |             |
| Port                                                                                                                                                        | Hosts       |
| 0 / udp                                                                                                                                                     | 172.16.8.11 |

UDP не найдены.

3. На сервере S2 сменим номер порта для службы FTP на 25, номер порта для службы SMTP на 21, сменим приветственные сообщения службы FTP.

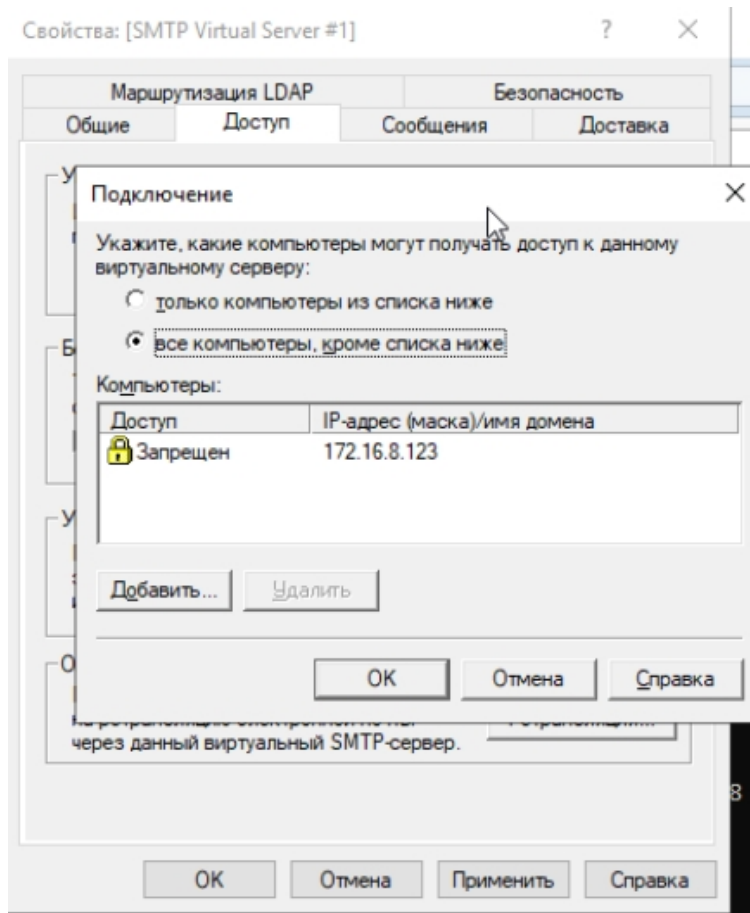


4. С помощью инструмента telnet проверим состояние портов 21 и 25 узла S2.

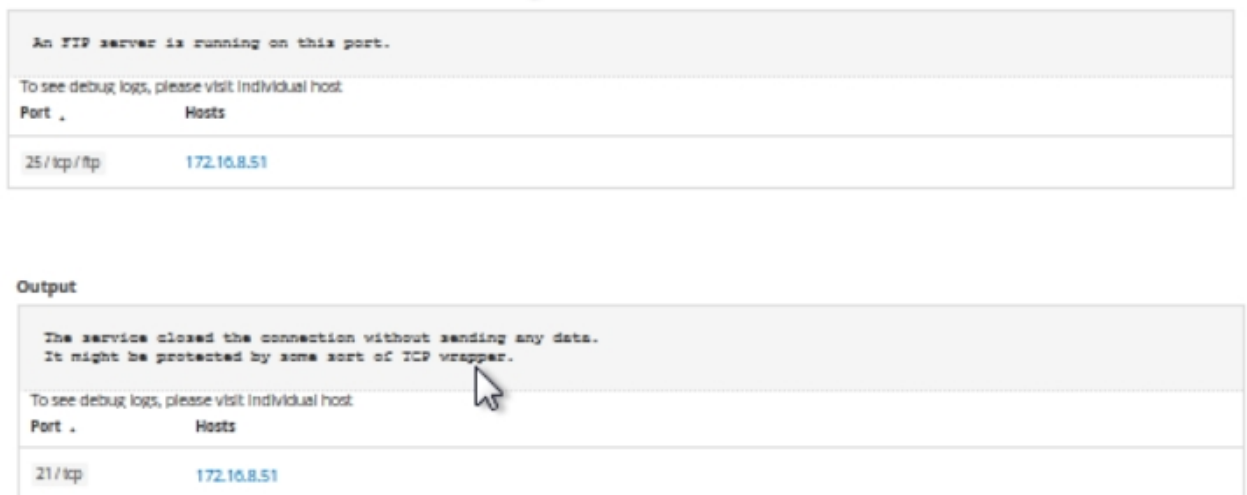
```
(user@kali)-[~]
$ telnet 172.16.8.51 25
Trying 172.16.8.51...
Connected to 172.16.8.51.
Escape character is '^]'.
220-Microsoft FTP Service
220 banner greeting
^]
telnet>
Connection closed.

(user@kali)-[~]
$ telnet 172.16.8.51 21
Trying 172.16.8.51...
Connected to 172.16.8.51.
Escape character is '^]'.
220 WIN-LHLAFV09N2C Microsoft ESMTS MAIL Service, Version: 10.0.17763.1697 ready at Sat, 14 Mar 2026 05:42:47 +0300
^]
```

5. Ограничим на узле S2 доступ к SMTP с узла TWS2.

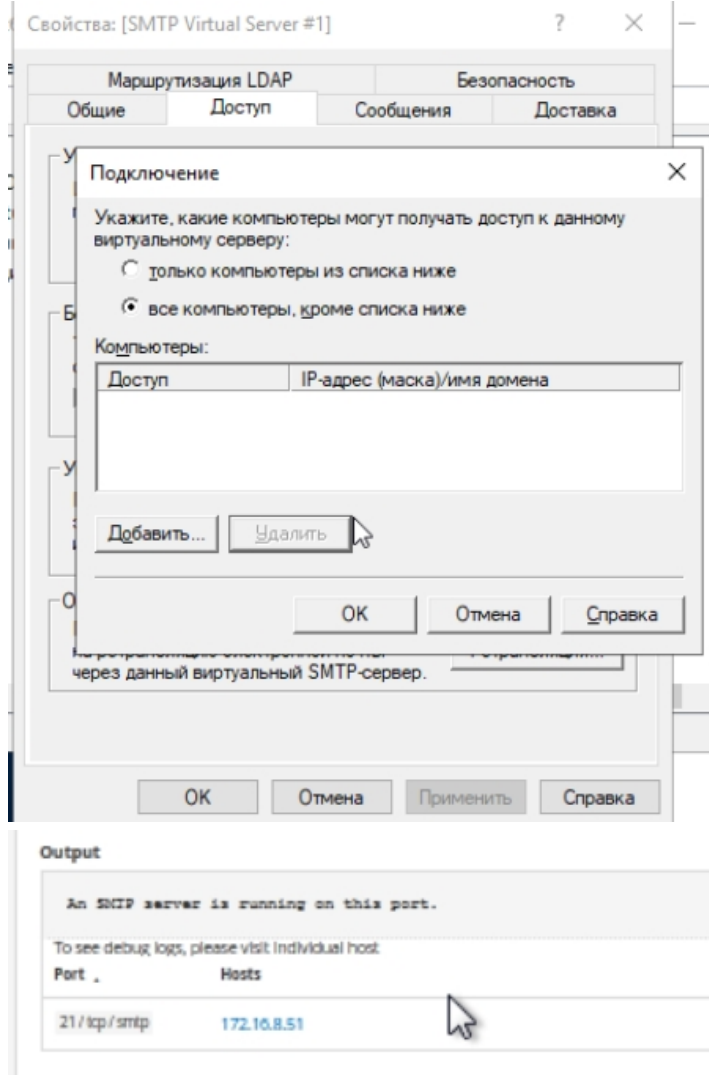


С помощью Nessus проверим результат



Как видим, определить SMTP на 21 порту не удалось.

## 6. Разрешим доступ к SMTP и повторим сканирование



## 7. На TWS1 с помощью nmap и amap выполним идентификацию служб и приложений узлов S1 и S2.

```
(user@kali)-[~]
$ nmap -sV 172.16.8.51
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-13 23:14 EDT
Nmap scan report for 172.16.8.51
Host is up (0.017s latency).
Not shown: 985 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  smtp         Microsoft ESMTMP 10.0.17763.1697
22/tcp    open  ssh         OpenSSH for_Windows_9.5 (protocol 2.0)
25/tcp    open  ftp         Microsoft ftpd
53/tcp    open  domain      Simple DNS Plus
80/tcp    open  http        Microsoft IIS httpd 10.0
110/tcp   open  pop3        MailEnable POP3 Server
135/tcp   open  msrpc       Microsoft Windows RPC
139/tcp   open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
1025/tcp  open  smtp        MailEnable smptd 0-1.87-
1051/tcp  open  ftp         FileZilla ftpd 0.9.4d beta
1521/tcp  open  oracle-tns  Oracle TNS listener 19.0.0.0.0 (unauthorized)
3389/tcp  open  ms-wbt-server Microsoft Terminal Services
5357/tcp  open  http        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp  open  http        Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 50:1C:E8:00:03:00 (Unknown)
Service Info: Hosts: WIN-LHLAFV09N2C, WIN-LHLAFV09N2C.example.com; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.24 seconds
```

```
(user@kali)-[~]
$ amap 172.16.8.51 21
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-13 23:15:26 - APPLICATION MAPPING mode

Protocol on 172.16.8.51:21/tcp matches smtp

Unidentified ports: none.

amap v5.4 finished at 2026-03-13 23:15:27
```

```
(user@kali)-[~]
$ amap 172.16.8.51 22
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-13 23:15:44 - APPLICATION MAPPING mode

Protocol on 172.16.8.51:22/tcp matches ssh
Protocol on 172.16.8.51:22/tcp matches ssh-openssh

Unidentified ports: none.

amap v5.4 finished at 2026-03-13 23:15:45
```

```
(user@kali)-[~]
$ amap 172.16.8.51 25
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-13 23:16:15 - APPLICATION MAPPING mode

Protocol on 172.16.8.51:25/tcp matches ftp

Unidentified ports: none.

amap v5.4 finished at 2026-03-13 23:16:15
```

```
(user@kali)-[~]
$ amap 172.16.8.51 53
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-13 23:16:33 - APPLICATION MAPPING mode

Unidentified ports: 172.16.8.51:53/tcp (total 1).

amap v5.4 finished at 2026-03-13 23:16:39
```

```
(user@kali)-[~]
$ amap 172.16.8.51 80
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-13 23:16:44 - APPLICATION MAPPING mode

Protocol on 172.16.8.51:80/tcp matches http
Protocol on 172.16.8.51:80/tcp matches http-apache-2
Protocol on 172.16.8.51:80/tcp matches http-iis

Unidentified ports: none.

amap v5.4 finished at 2026-03-13 23:16:44
```

nmap -sV 172.16.8.11

```
Not shown: 983 filtered tcp ports (no-response), 11 filtered tcp ports
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 3.0.5
22/tcp    open  ssh          OpenSSH 9.9 (protocol 2.0)
25/tcp    open  smtp?
53/tcp    open  domain       ISC BIND 9.18.41
80/tcp    open  http         Apache httpd 2.4.65 ((Fedora Linux))
9090/tcp   open  ssl/http     Cockpit web service
1 service unrecognized despite returning data. If you know the service
```

```
(user@kali)-[~]
$ amap 172.16.8.11 21
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-14 02:08:16 - APPLICATION MAPPING mode

Protocol on 172.16.8.11:21/tcp matches ftp

Unidentified ports: none.

amap v5.4 finished at 2026-03-14 02:08:16
```

```
(user@kali)-[~]
$ amap 172.16.8.11 22
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-14 02:08:40 - APPLICATION MAPPING mode

Protocol on 172.16.8.11:22/tcp matches ssh
Protocol on 172.16.8.11:22/tcp matches ssh-openssh

Unidentified ports: none.

amap v5.4 finished at 2026-03-14 02:08:40
```

```
(user@kali)-[~]
$ amap 172.16.8.11 25
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-14 02:09:07 - APPLICATION MAPPING mode

Unidentified ports: 172.16.8.11:25/tcp (total 1).

amap v5.4 finished at 2026-03-14 02:09:13
```

```
(user@kali)-[~]
$ amap 172.16.8.11 53 80
amap v5.4 (www.thc.org/thc-amap) started at 2026-03-14 02:09:48 - APPLICATION MAPPING mode

Protocol on 172.16.8.11:80/tcp matches http
Protocol on 172.16.8.11:80/tcp matches http-apache-2
Protocol on 172.16.8.11:53/tcp matches dns
```

8. С помощью swaks выполним идентификацию SMTP узлов S1 и S2.

```
(user@kali)-[~]
$ swaks
To: 172.16.8.11
== Trying 172.16.8.11:25 ...
== Connected to 172.16.8.11.
EHLO kali
< ** Timeout (30 secs) waiting for server response
-> QUIT
^C
```



```

(user@kali)-[~]
$ swaks
To: 172.16.8.51:21
== Trying 172.16.8.51:21:25 ...
== Connected to 172.16.8.51:21.
<- 220 WIN-LHLAFV09N2C Microsoft ESMTMP MAIL Service, Version: 10.0.17763.1697 ready at Sat, 14 Mar 2026 09:21:00 +0300
-> EHLO kali
<- 250-WIN-LHLAFV09N2C Hello [172.16.8.124]
<- 250-AUTH=LOGIN
<- 250-AUTH LOGIN
<- 250-TURN
<- 250-SIZE 2097152
<- 250-ETRN
<- 250-PIPELINING
<- 250-DSN
<- 250-ENHANCEDSTATUSCODES
<- 250-8bitmime
<- 250-BINARYMIME
<- 250-CHUNKING
<- 250-VRFY
<- 250 OK
-> MAIL FROM:<user@kali>
<- 250 2.1.0 user@kali...Sender OK
-> RCPT TO:<172.16.8.51:21>
<-- 501 5.5.4 Invalid Address
-> QUIT
<- 221 2.0.0 WIN-LHLAFV09N2C Service closing transmission channel
== Connection closed with remote host.

```

9. С помощью telnet выполним идентификацию HTTP узлов S1, S2

```

(user@kali)-[~]
$ telnet 172.16.8.51 80
Trying 172.16.8.51...
Connected to 172.16.8.51.
Escape character is '^]'.
GET
HTTP/1.1 400 Bad Request
Content-Type: text/html; charset=us-ascii
Server: Microsoft-HTTPAPI/2.0
Date: Sat, 14 Mar 2026 06:22:18 GMT
Connection: close
Content-Length: 326

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<HTML><HEAD><TITLE>Bad Request</TITLE>
<META HTTP-EQUIV="Content-Type" Content="text/html; charset=us-ascii"></HEAD>
<BODY><h2>Bad Request - Invalid Verb</h2>
<hr><p>HTTP Error 400. The request verb is invalid.</p>
</BODY></HTML>
Connection closed by foreign host.

```

```

(user@kali)-[~]
$ telnet 172.16.8.11 80
Trying 172.16.8.11...
Connected to 172.16.8.11.
Escape character is '^]'.
GET
HTTP/1.1 400 Bad Request
Date: Sat, 14 Mar 2026 06:23:27 GMT
Server: Apache/2.4.65 (Fedora Linux)
Content-Length: 226
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
</body></html>
Connection closed by foreign host.

```